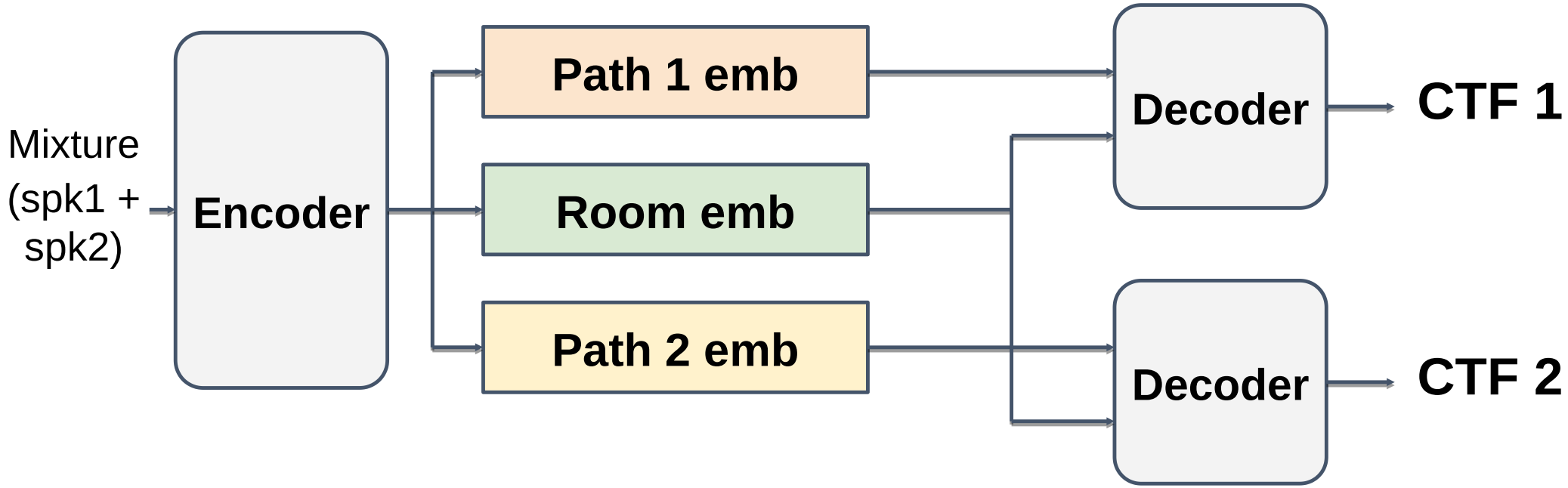
CTF(RIR) Estimator: Room / Path Learning

Learn acoustic features that remain consistent when speakers and utterances change.



Shared decoder weights

Two views → the same encoder and decoder

View A: original speakers

View B: different speakers

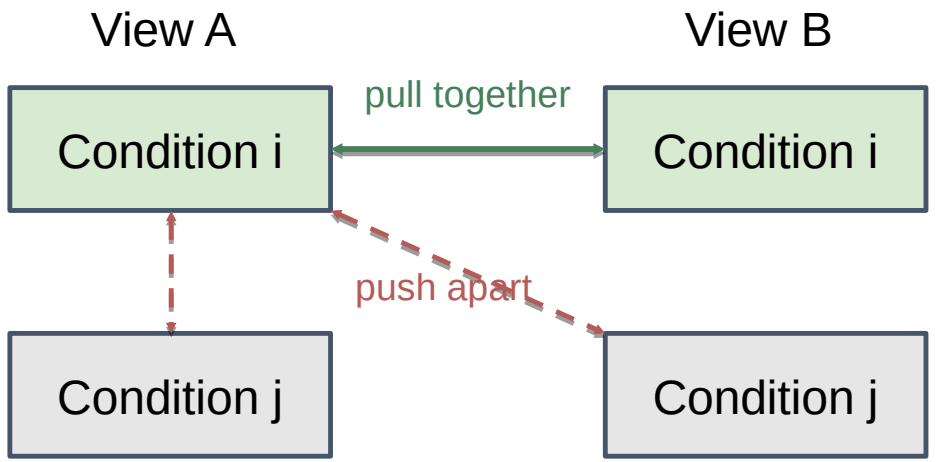
Fixed room, microphone, and RIR pair
Add Lroom and Lpath during training

Same physical paths across views; room/path labels denote the feature-sharing structure.

Contrastive Losses: Lroom and Lpath

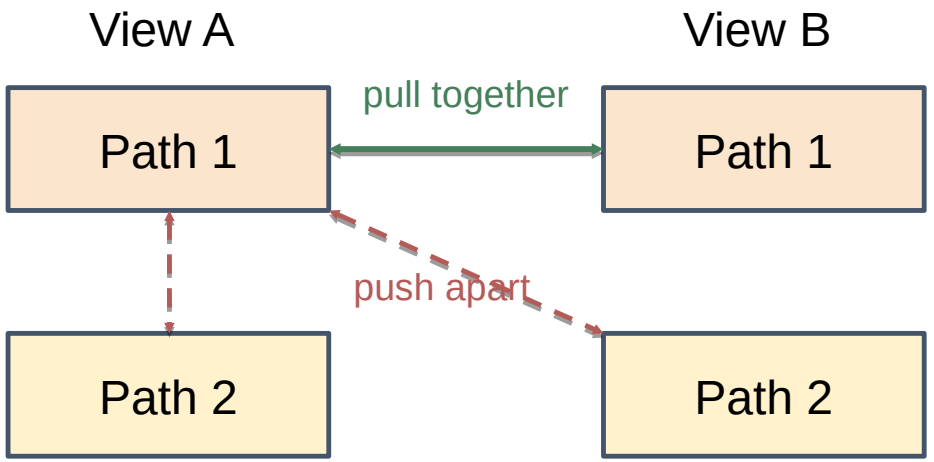
Positive: same acoustic condition across views Negative: a different condition / path

Lroom: shared room features



+ Same fixed RIR pair, different speech
− Other batch items (both views)

Lpath: path-specific features



+ Same physical path, aligned by CTF PIT
− The other path in the same item only

Symmetric InfoNCE | Cosine similarity | Temperature $\tau = 0.1$

$$L_{total} = \tfrac{1}{2}(L_{sweep}^{A} + L_{sweep}^{B}) + 0.01\, L_{room} + 0.01\, L_{path}$$

Training-only projections: 64 → 32 per frequency. Lpath uses path codes before slot embeddings.